

Chapter 8: Gases and Gas Laws.

The first substances to be produced and studied in high purity were gases. Gases are more difficult to handle and manipulate than solids and liquids, since any minor mistakes generally results in the gas escaping to the atmosphere. However, the ability to produce gases in very high purity made the additional difficulty acceptable. The most common way of producing a gas was by some sort of chemical reaction, and the gas was collected by liquid displacement (either water or mercury). Figure 8.1 shows the general process of collecting gas by liquid displacement.

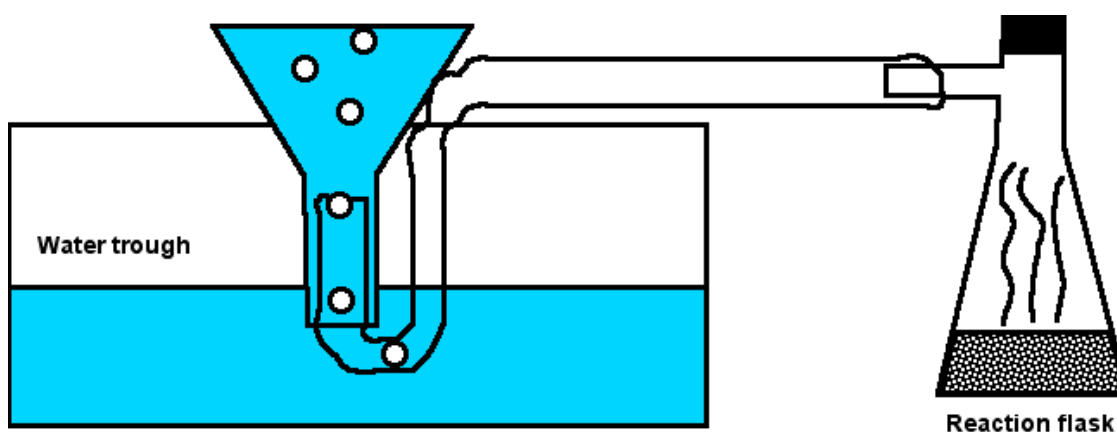


Figure 8.1. Method of displacement for collecting gases.

A water-filled container is inverted and placed into a water trough. A rubber hose is placed in the mouth of the container, with the other end attached to a reaction flask. The chemical reaction produces gas, which flows through the tube and displaces water from the container. By selecting the proper reactant masses, sufficient gas to fill 3 – 6 containers can be produced. Typically, the first container collected isn't saved, since it contains residual air from the reaction flask.

Using this general method, scientists produced and characterized hydrogen, oxygen, nitrogen, carbon dioxide, sulfur dioxide, chlorine, and several other gases. Once they obtained reasonably pure gases, systematic experimentation led to other discoveries.

Generally, gases have properties substantially different than solids or liquids. Gases do not have fixed volumes; instead their volume depends directly upon pressure and temperature. Gases don't have a fixed shape, but are said to "take the shape of their container". Gases do have a fixed mass, although measuring the mass may be difficult sometimes.

Boyle's law.

In 1662, English natural philosopher Robert Boyle (1627 – 1691) published what is now called Boyle's law – the product of a gas' pressure and volume is constant:

$$P \times V = k$$

Gases therefore show an inverse relationship between pressure and volume; as pressure increases, volume decreases and vice versa. For Boyle's law to be obeyed, the temperature has to remain constant.

Gas pressure are measured using a variety of units. Commonly encountered units are given in Table 8.1.

14.7 pounds per square inch (psi)	= 1 atmosphere (atm)
	= 760 mmHg
	= 760 Torr
	= 29.92 inches Mercury
	= 33.9 Feet of water
	= 101,325 Pascal (Pa)

Table 8.1. Pressure values and equivalents.

The weight of the atmosphere at sea level equals 14.7 pounds per square inch, and this is defined as 1 standard atmosphere of pressure. The average man has about 2945 in² total body surface area, while the average woman has about 2480 in². The total pressure exerted by the atmosphere is 21.6 tons for men and 18.2 tons for women.

One common method of measuring air pressure is by determining the height of a column of liquid supported by air pressure. This method was first used by Italian physicist and mathematician, Evangelista Torricelli (1608 – 1647). Torricelli was a colleague of Galileo, and both scientists were trying to solve an important practical problem. Pump makers were unable to build suction-type water pumps (Figure 8.2) that could raise water higher than about 10 meters. Galileo believed that the pumps were poorly built, but Torricelli had a different idea.

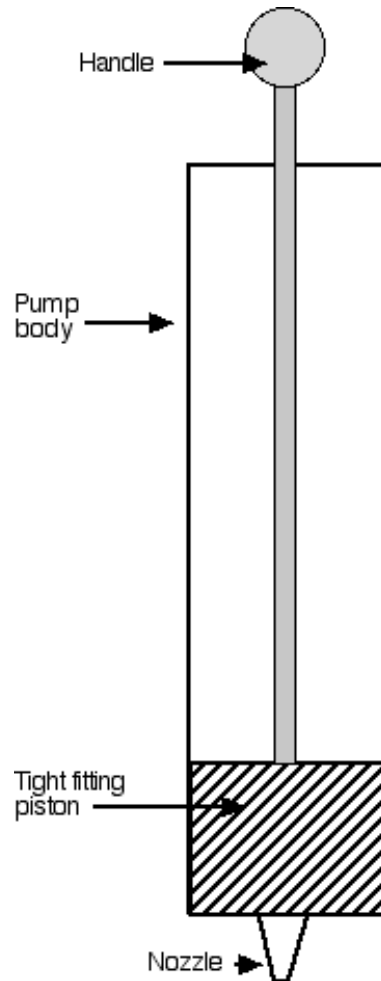


Figure 8.2. Simple suction pump. Pulling the handle out raises the piston and draws water into the pump body. Pushing the handle in lowers the piston and forces water out of the pump.

Torricelli experimented with tubes filled with water. He inverted these tubes over a container of water, discovering that a very long column of water could be maintained above the surface of water in the container. Torricelli realized that the liquid column would be inconveniently tall if he continued using water, and switched to mercury. When he used mercury in his tubes, the mercury in the tube fell a small distance and stopped (Figure 8.3).

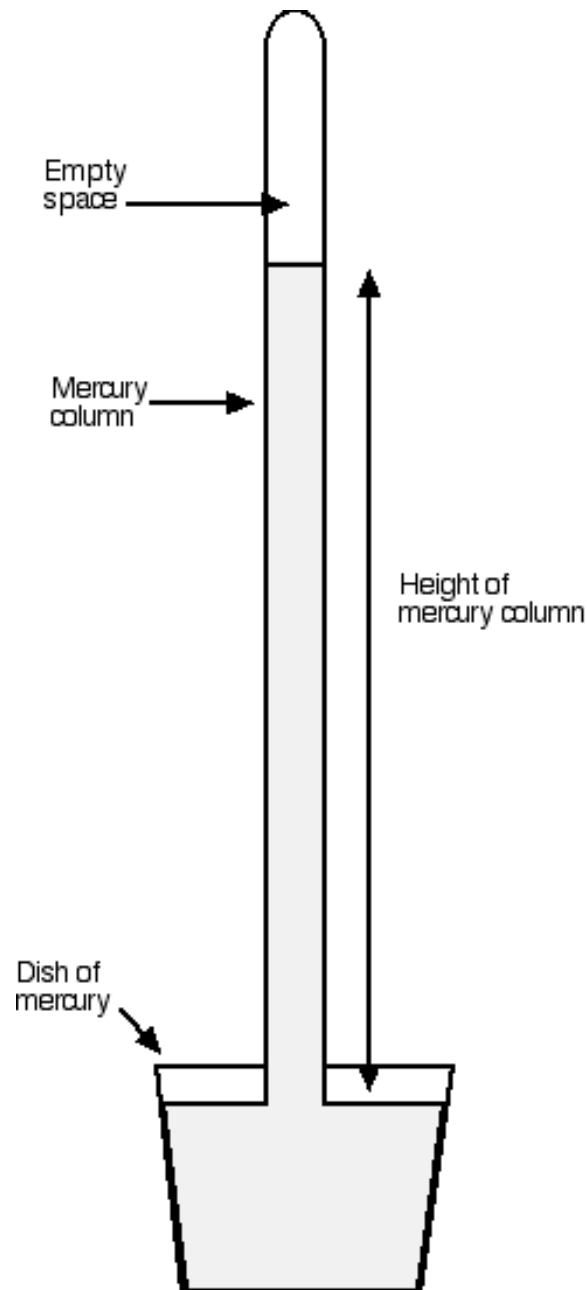


Figure 8.3. Torricelli's barometer.

Torricelli realized that the empty space in the tube was truly empty – a vacuum, now called a ***Torricellian vacuum*** in his honor. Torricelli calculated the weight of the mercury column, and the area of the column in contact with the surface of the mercury dish. The pressure exerted by the column of mercury was 14.7 psi. If the mercury column was exerting a pressure of 14.7 psi downward, then an exactly equal force must be exerted upward on the mercury column. If the forces weren't equal, then either the empty space in the tube would fill with mercury, or

the mercury would flow out of the tube. This balancing force must come from the air: air presses down on the mercury in the dish, the force is transferred to the mercury at the bottom of the column, and is equal to the weight of mercury in the tube.

The unit “Torr” is named in Torricelli’s honor, and is equivalent to 1 mmHg.

The metric unit of pressure uses metric units of force (Newtons, N) and area (square meters, m²). 1 N/m² equals 1 Pascal (Pa), named for French physicist Blaise Pascal (1623 – 1662), who conducted pioneering experiments in hydraulics and hydrostatics.

Amontons’ law.

In the late 1600’s, French physicist Guillaume Amontons (1663 – 1705) investigated the relationship between temperature and pressure. Although his work was not very quantitative, it did point the way towards the idea of absolute zero. Amontons found that the pressure of a gas divided by temperature was equal to a constant. When the gas pressure was zero (the lowest pressure you can achieve), the equivalent temperature would be zero. Amontons’ law commonly has the form:

$$\frac{P}{T} = k$$

In all gas calculations, we use Kelvin temperature (°C + 273 = K) to avoid problems with negative values for Celsius or Fahrenheit temperatures.

Charles’ law.

In 1783, French balloonist, inventor, and scientist Jacques Alexandre César Charles (1746 – 1832) used a hydrogen-filled balloon to ascend to an altitude of 3000 feet. In 1787, he noted in a general way that changes in pressure and temperature affected the volume of a gas.

As a balloonist, Charles was very interested in the properties of gases. However, Charles didn’t clearly recognize anything approaching a natural law, nor did he produce any equation summarizing his observations. He did not provide any written description of his experiments, nor did he present any experimental data indicating he systematically studied the effect of temperature on pressure or volume.

He did communicate his general observations to French physicist and chemist Joseph Louis Gay-Lussac (1778 – 1850). Gay-Lussac systematically studied the effects of temperature on volume, maintaining a constant pressure during his

experiments. He also investigated the temperature – volume relationship with several different gases. In his 1802 paper describing his experimental results, he acknowledged Charles' preliminary work describing the expansion of gases over a temperature range of 0 to 80 °C. Gay-Lussac's work became known as "Charles' Law":

$$\frac{V}{T} = k$$

If the temperature is measured in Kelvin, then a change of 1 K results in a volume change equal to 1/273 of the original volume. Imagine we have 1.00000 L of gas at 298 K (25 °C). We raise the temperature to 299 K. The volume increases to 1.003 66 L (1/273 = 0.003 66). To double the volume of gas, the temperature must rise to 596 K (323 °C). This relationship between volume change and temperature complimented Amontons' law, but Gay-Lussacs experiments were much more quantitative than were Amontons'.

Frequently, Amontons' and Charles' laws are combined together to produce the combined gas law:

$$\frac{P \times V}{T} = k$$

We often use the combined gas law to calculate equivalent values at standard temperature and pressure (STP). Standard pressure is 1.00 atmosphere, and standard temperature is 0 °C (273 K). To make this conversion we recognize that any three values for pressure, volume and temperature must give us the same value for the constant k . We can then write the equation:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \left(\text{or } \frac{P_i V_i}{T_i} = \frac{P_f V_f}{T_f} \right)$$

In the alternate version, i and f indicate initial and final sets of P,V, and T.

Kinetic molecular theory.

The gas laws of Boyle, Charles, Amontons, and Gay-Lussac eventually led to the development of the kinetic molecular theory. Although initially applied just to gases, the theory was eventually extended to solids and liquids. There are several postulates to the theory:

1. Gases are composed of tiny particles (atom or molecules) in constant motion.

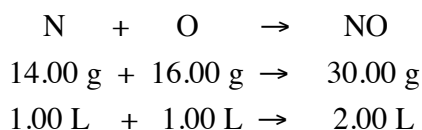
2. The distance between individual particles is relatively large compared to the size of the gas particles.
3. There are virtually no interactions between the individual gas particles, other than collisions. During these collisions, no chemical reactions occur – just the exchange of kinetic energy.
4. Temperature is directly related to kinetic energy and therefore velocity. The average kinetic energy is equal to $3kT/2$, where k is the Boltzmann constant (1.38×10^{-23} J/K).

Avogadro's law.

In 1811 an obscure Italian physicist, Lorenzo Romano Amedeo Carlo Avogadro (1776 – 1856) wrote *“Essay on Determining the Relative Masses of the Elementary Molecules of Bodies”* in which he proposed that for constant temperature and pressure, equal volumes of two different gases contained identical numbers of particles. If we set the number of particles equal to n , then Avogadro's law becomes:

$$\frac{V}{n} = k$$

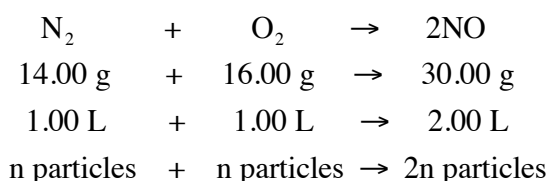
By this time, chemists had made numerous measurements of the compositions of gases such as nitric oxide (NO), and accepted the idea that elements combine in fixed proportions by weight and by volume, with these combinations giving the same chemical formula. However, there were difficulties with this idea. Some chemists used the terms “atom”, “molecule”, “atomic weight”, “molecular weight”, and “equivalent weight” interchangeably. This led to significant confusion in their calculations and in their understanding of exactly what was happening in chemical reactions. Consider the reaction between nitrogen and oxygen to make nitric oxide:



When 14.00 grams of nitrogen combines with 16.00 grams of oxygen, 30.00 grams of nitric oxide forms. Therefore, the simplest formula for nitric oxide is NO, containing one atom each of nitrogen and oxygen. If equal volumes of gases contain equal number of atoms, then 1.00 liter of nitrogen should combine with 1.00 liter of oxygen to produce 1.00 liter of nitric oxide. However, this isn't what happens – instead 2.00 liters of nitric oxide are produced!

This same problem happens when hydrogen combines with chlorine to make hydrogen chloride ($\text{H} + \text{Cl} \rightarrow \text{HCl}$), or when hydrogen combines with oxygen to produce water ($\text{H} + \text{O} \rightarrow \text{HO}$!).

The underlying problem was the belief that gases were simple individual atoms (although some problems were traceable to misunderstandings of the chemical formula of compounds, such as HO for water). Avogadro didn't believe that all gases were necessarily single atoms. If nitrogen and oxygen were N_2 and O_2 , then:



The ratio of nitrogen/oxygen is still 14.00/16.00 (7/8), so the mass composition values are consistent, and now the combining volumes make sense and are also consistent with the mass values.

Avogadro's controversial idea wasn't generally accepted until after 1860.

Ideal Gas Law.

Combining all of the other gas laws together produces the Ideal Gas Law:

$$PV = nRT$$

P , V , and T are pressure, volume, and temperature, n is the number of particles, and R is the gas constant. While R can have a variety of values depending on units, the most commonly used value is 0.0821 L-atm/mol-K. Since all other gas laws were combined to produce the Ideal Gas Law, it shouldn't surprise you that all other gas laws can be obtained from it.

If I have a fixed number of moles of gas (if my gas is in a closed container for example), then n and R are constant, and the (mathematical) product of two constants is a constant. Then the Ideal Gas Law becomes:

$$\begin{aligned}
 PV &= kT \\
 \frac{PV}{T} &= k
 \end{aligned}$$

which is the combined gas law.

If we have fixed n and T , then nrT is constant and we get Boyle's law. If n and P are fixed, then our equation becomes Charles' law, while if n and V are fixed, we get Amontons' law. If we fix V and P , then we get Avogadro's law. The only gas law you need is the Ideal Gas Law, and a little bit of algebra.

Dalton's law of partial pressures.

One of the most interesting characters in chemical history is the Quaker chemist, physicist, and meteorologist John Dalton (1766 – 1844). Dalton is known not only for his research into atomic theory, but also his study of colorblindness, once called *Daltonism*: Dalton was colorblind. In developing his atomic theory, Dalton relied heavily on years of meteorology measurements to help guide his understanding of the atmosphere. To aid his analysis, Dalton drew simple sketches (Figure 8.4) helping him visualize how atmospheric gases mixed.

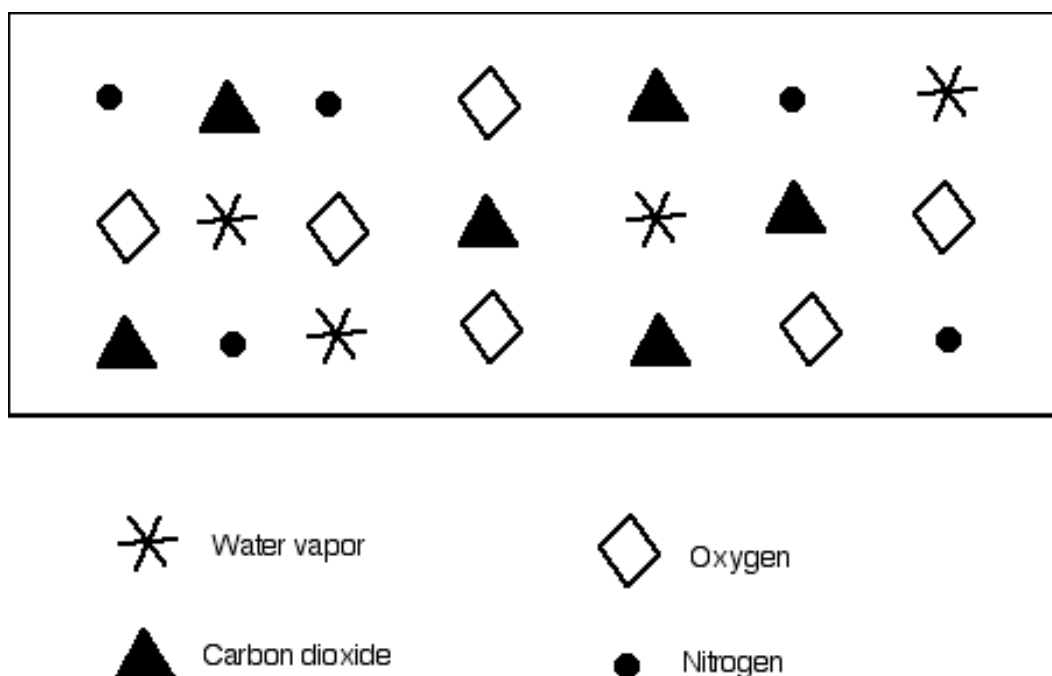


Figure 8.4. Dalton's representation of gases in the atmosphere.

In 1801, Dalton stated his Law of Partial Pressure: the total pressure of a gas mixture is the sum of the individual pressures exerted by each gas in the mixture. This law can be easily understood by reference to Figure 8.4. The total pressure is due to the total number of individual atoms in the gas (I am using the term "atom" as Dalton understood it – an atom was the smallest particle of a particular substance. Some of Dalton's atoms were actually molecules!). The partial pressure of water vapor is proportional to the relative amount of water vapor in the gas mixture; the partial pressure of oxygen is proportional to the relative amount of oxygen in the